TENSORING A PERFECT COMPLEX WITH A NICE COMPLEX PUSHES FORWARD TO A PERFECT COMPLEX WITH BASE-CHANGE COMPATIBLE FORMATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: the Noetherian case is quoted and the general case descends along a filtered colimit

Editorial note. The label lemma-base-change-tensor-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the base-change compatibility of the total direct image of a tensor product (base-change-tensor lemma)
- the tensor-perfect lemma over Noetherian bases
- Noetherian approximation: descent of finite presentation, modules, flatness and properness of support along filtered colimits of rings
- the perfect-on-limit lemma descending perfect objects

Source reference. Original reference 0A1D: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5625–5625. Prerequisite pointer invoked at proof line(s) 6321; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09RJ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6229–6229. Prerequisite pointer invoked at proof line(s) 6347; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09J5: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5794–5794. Prerequisite pointer invoked at proof line(s) 6324; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07SK: sources/stacks-chow/tags.txt, source lines 9905–9905. Prerequisite pointer invoked at proof line(s) 6332; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08K0: sources/stacks-chow/tags.txt, source lines 10868–10868. Prerequisite pointer invoked at proof line(s) 6343; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07V7: sources/stacks-chow/tags.txt, source lines 9997–9997. Prerequisite pointer invoked at proof line(s) 6336; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08K2: sources/stacks-chow/tags.txt, source lines 10870–10870. Prerequisite pointer invoked at proof line(s) 6346; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of finite presentation between algebraic spaces over S. Let $E \in D(\mathcal{O}_X)$ be a perfect object. Let $\mathcal{G}^\bullet$ be a bounded complex of finitely presented $\mathcal{O}_X$-modules, flat over Y, with support proper over Y. Then*

$$K = Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$$

is a perfect object of $D(\mathcal{O}_Y)$ and its formation commutes with arbitrary base change.

Proof. The statement on base change is Lemma 0A1D. Thus it suffices to show that K is a perfect object. If Y is Noetherian, then this follows from Lemma 09J5. We will reduce to this case by Noetherian approximation. We encourage the reader to skip the rest of this proof.

The question is local on Y, hence we may assume Y is affine. Say $Y = \mathrm{Spec}(R)$. We write $R = \mathrm{colim}\, R_i$ as a filtered colimit of Noetherian rings R_i. By Limits of Spaces, Lemma 07SK there exists an i and an algebraic space X_i of finite presentation over R_i whose base change to R is X. By Limits of Spaces, Lemma 07V7 we may assume after increasing i, that there exists a bounded complex of finitely presented $\mathcal{O}_{X_i}$-modules $\mathcal{G}_i^\bullet$ whose pullback to X is $\mathcal{G}^\bullet$. After increasing i we may assume $\mathcal{G}_i^n$ is flat over R_i, see Limits of Spaces, Lemma 08K0. After increasing i we may assume the support of $\mathcal{G}_i^n$ is proper over R_i, see Limits of Spaces, Lemma 08K2. Finally, by Lemma 09RJ we may, after increasing i, assume there exists a perfect object E_i of $D(\mathcal{O}_{X_i})$ whose pullback to X is E. By Lemma 09J5 we have that $K_i = Rf_{i,*}(E_i \otimes^{\mathbf{L}}_{\mathcal{O}_{X_i}} \mathcal{G}_i^\bullet)$ is perfect on $\mathrm{Spec}(R_i)$ where $f_i : X_i \to \mathrm{Spec}(R_i)$ is the structure morphism. By the base change result (Lemma 0A1D) the pullback of K_i to $Y = \mathrm{Spec}(R)$ is K and we conclude. □